

Department of Computer Science and Engineering
Islamic University of Technology (IUT)
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Lab Report 03

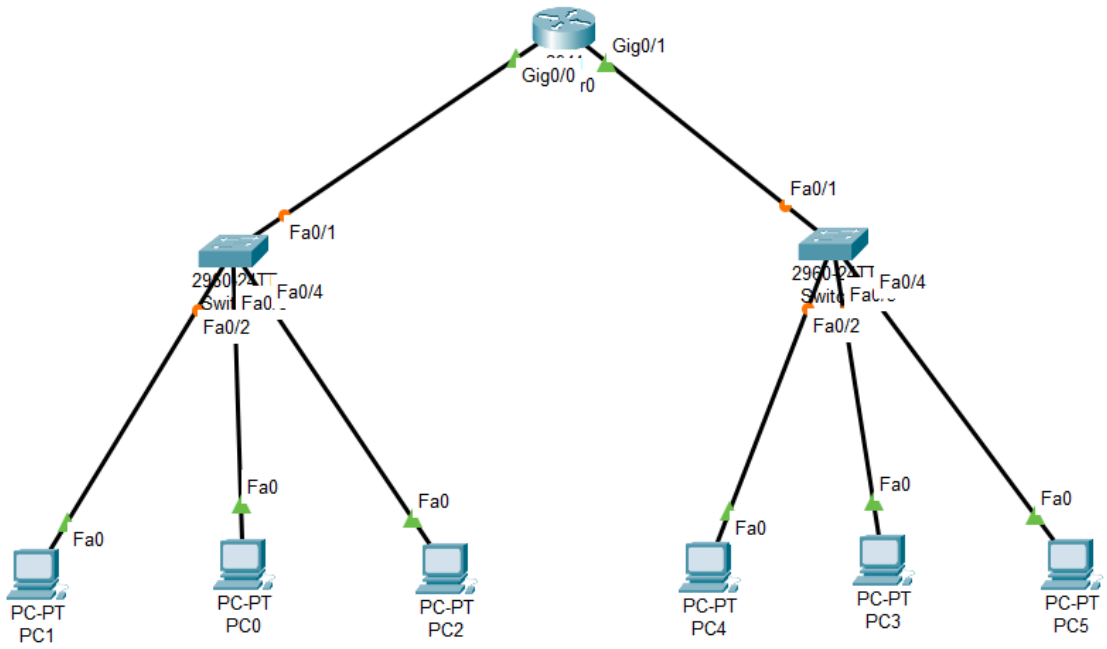
CSE 4412: Data Communication and Networking Lab

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Section: B
Semester: 4th
Academic Year: 2022-2023

Date of Submission:12/02/2024

Diagram of the experiment:

TASK #01:



As I can connect a maximum of 32 (last 2 digits of my ID) hosts per network, in the last octet of the Subnet we will need 5 bits for the host address (26=64), and the first 2 bits of the last octet will be part of the network address.

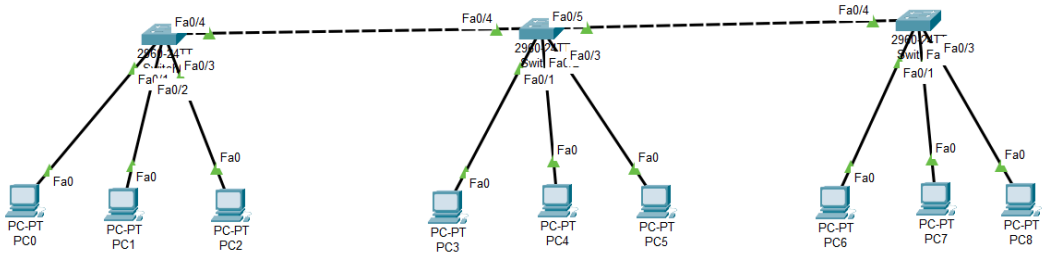
Subnet: 255.255.255.192
IP range of the first subnetwork: 192.168.42.1 – 192.168.42.64
Default gateway of subnetwork 1st subnetwork: 192.168.42.1
IP range of the second subnetwork: 192.168.52.1 – 192.168.52.64
Default gateway of subnetwork 2nd subnetwork: 192.168.52.1

Step 2: Router Configuration
To configure the router, I clicked on the router and went to CLI. Ther used the following commands:

Router# enable

Router# config t
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 196.168.42.1 255.255.255.192
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 196.168.52.1 255.255.255.192
Router(config-if)# no shutdown
Router(config-if)# exit

TASK #02:



Working Procedure:

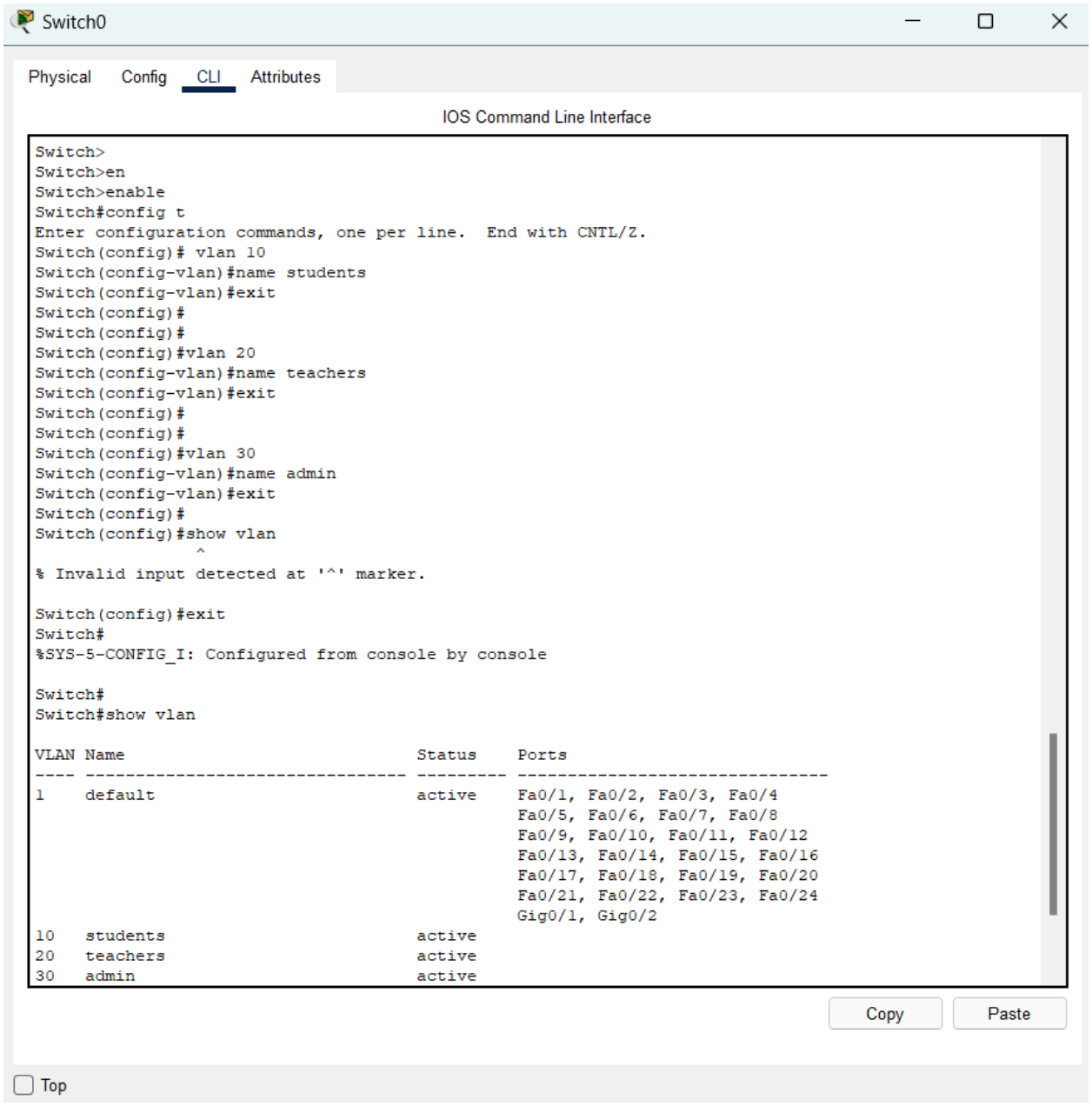
Step 1: Configuring Switches

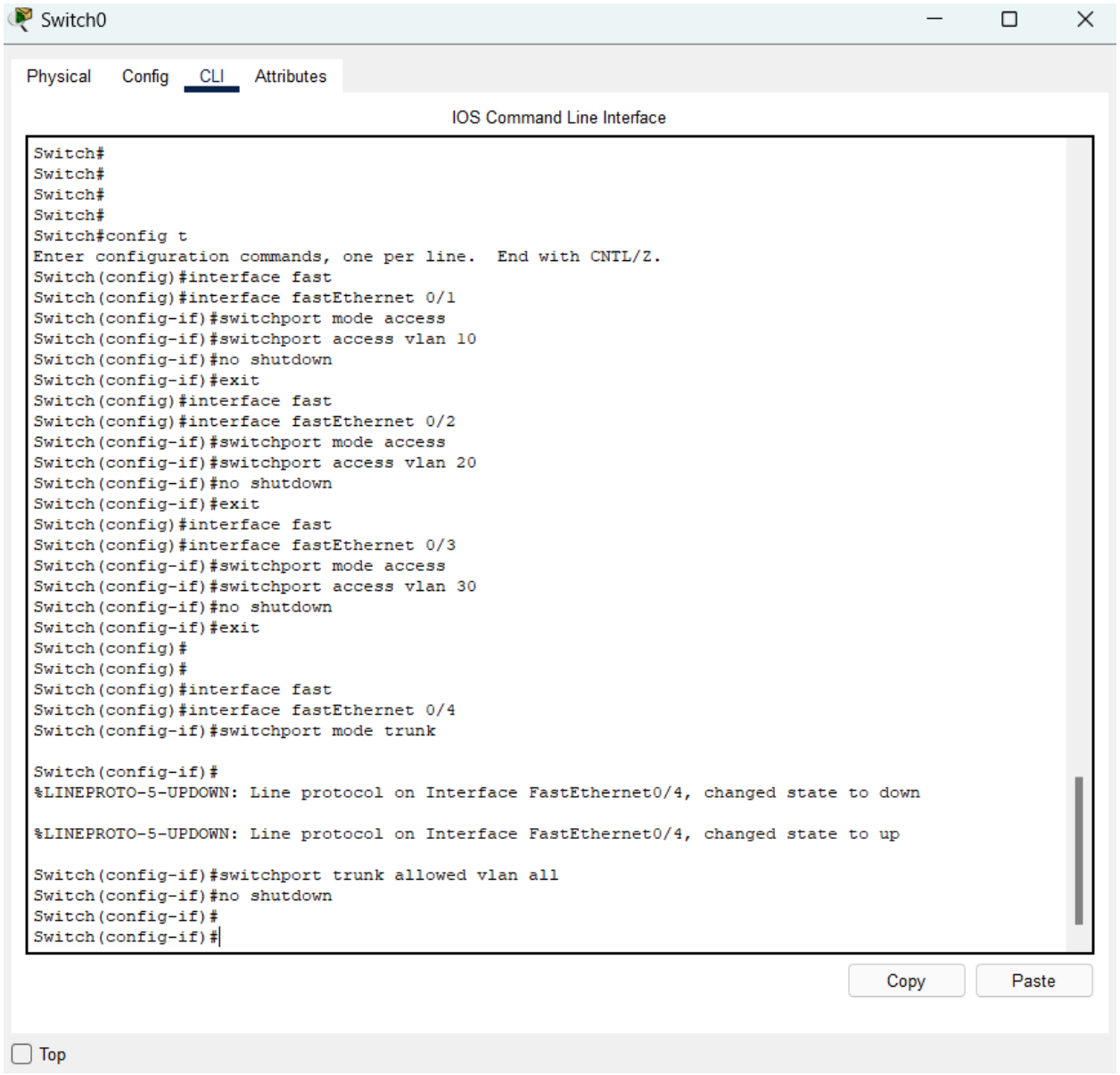
At first, I configured 3 VLANs with VLAN IDs 10, 20, and 30 inside the switch and assigned appropriate names. To do this, I have used the following commands:
Switch> enable
Switch# config t
Switch(config)# vlan 10
Switch(config-vlan)# name student
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name teacher
Switch(config-vlan)# exit
Switch(config)# vlan 30
Switch(config-vlan)# name admin
Switch(config-vlan)# exit

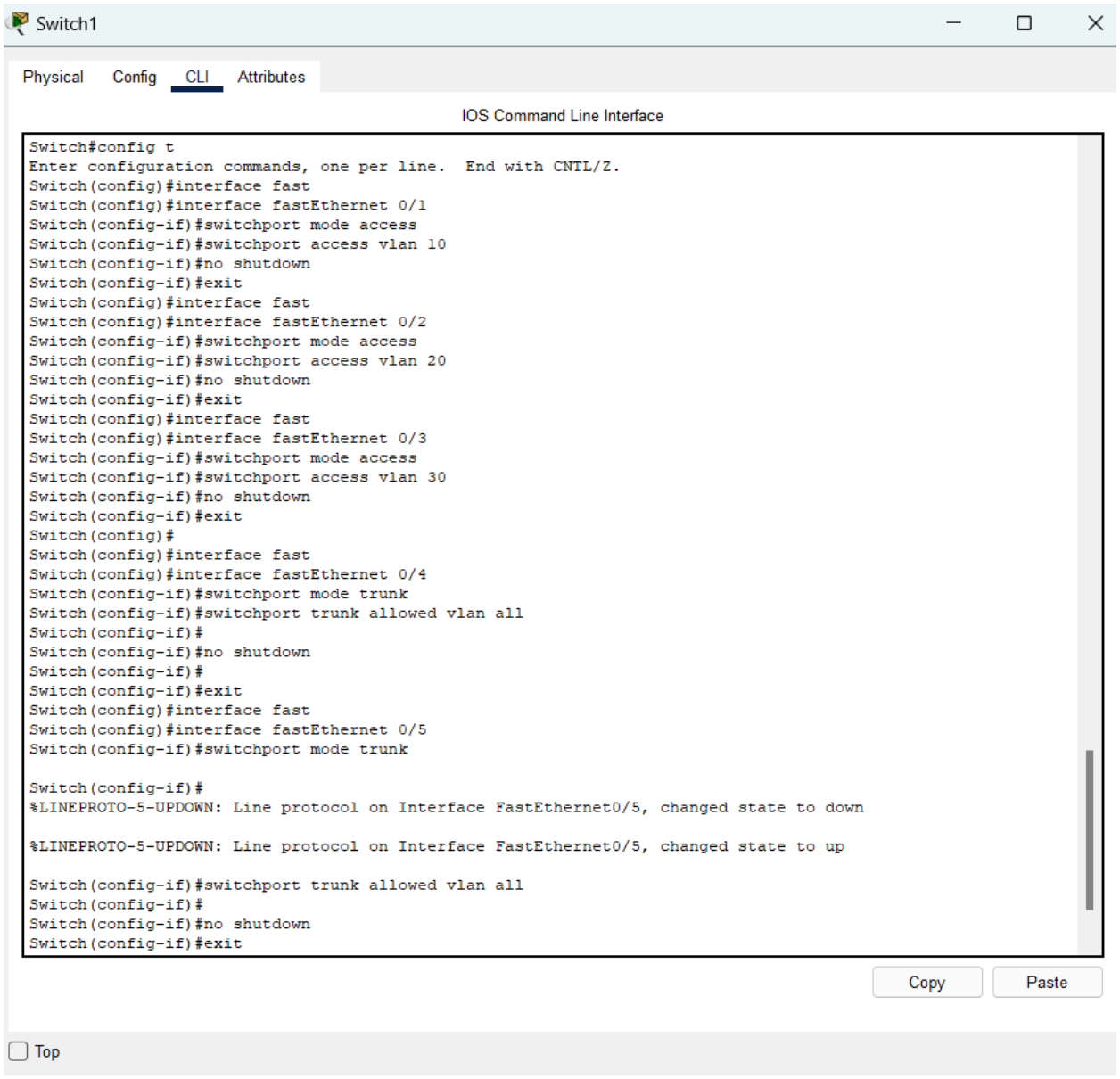
Then I configured the Interfaces belonging to each VLAN. For that, I used the following commands:

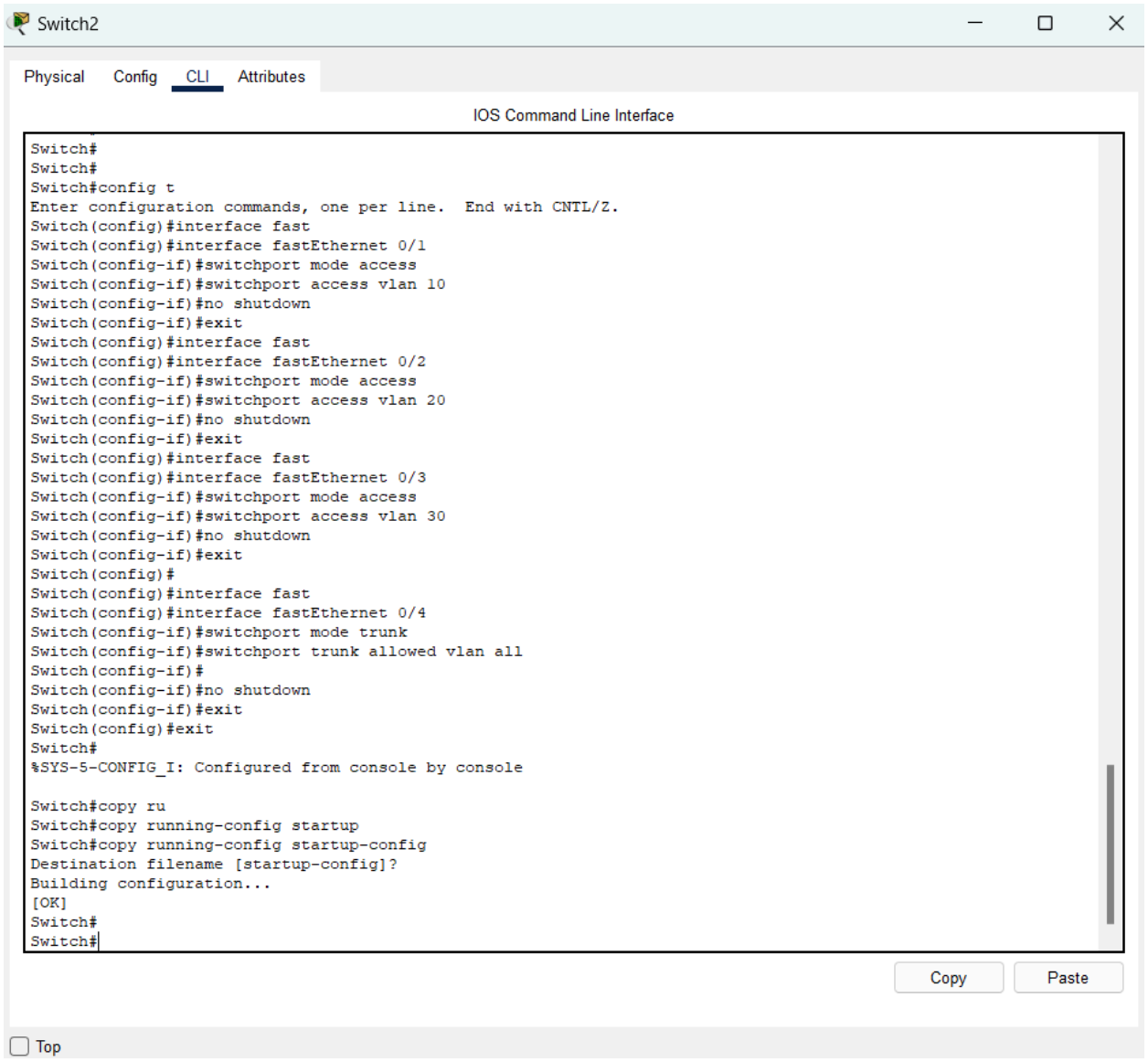
```
Switch(config)# interface fastEthernet 0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# no shutdown
Switch(config-if)# exit
Switch(config)# interface fastEthernet 0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# no shutdown
Switch(config-if)# exit
Switch(config)# interface fastEthernet 0/3
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
Switch(config-if)# no shutdown
Switch(config-if)# exit
Switch(config)# interface fastEthernet 0/4
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan all
Switch(config-if)# no shutdown
Switch(config-if)# exit
```

This way I have configured the interfaces. For other switches, the commands are exactly same. Just in switch 1 I truncked twice once with fastEthernet 0/4 and again with fastEthernet 0/5
Here are the screenshots of the three switches:


Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch>
Switch>en
Switch>enable
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)# vlan 10
Switch(config-vlan)#name students
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#
Switch(config)#vlan 20
Switch(config-vlan)#name teachers
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#
Switch(config)#vlan 30
Switch(config-vlan)#name admin
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#show vlan
^
% Invalid input detected at '^' marker.
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#
Switch#show vlan
VLAN Name Status Ports
---- -------------------------------- --------- -------------------------------
1 default active Fa0/1, Fa0/2, Fa0/3, Fa0/4
Fa0/5, Fa0/6, Fa0/7, Fa0/8
Fa0/9, Fa0/10, Fa0/11, Fa0/12
Fa0/13, Fa0/14, Fa0/15, Fa0/16
Fa0/17, Fa0/18, Fa0/19, Fa0/20
Fa0/21, Fa0/22, Fa0/23, Fa0/24
Gig0/1, Gig0/2
10 students active
20 teachers active
30 admin active
Copy
Paste
Top

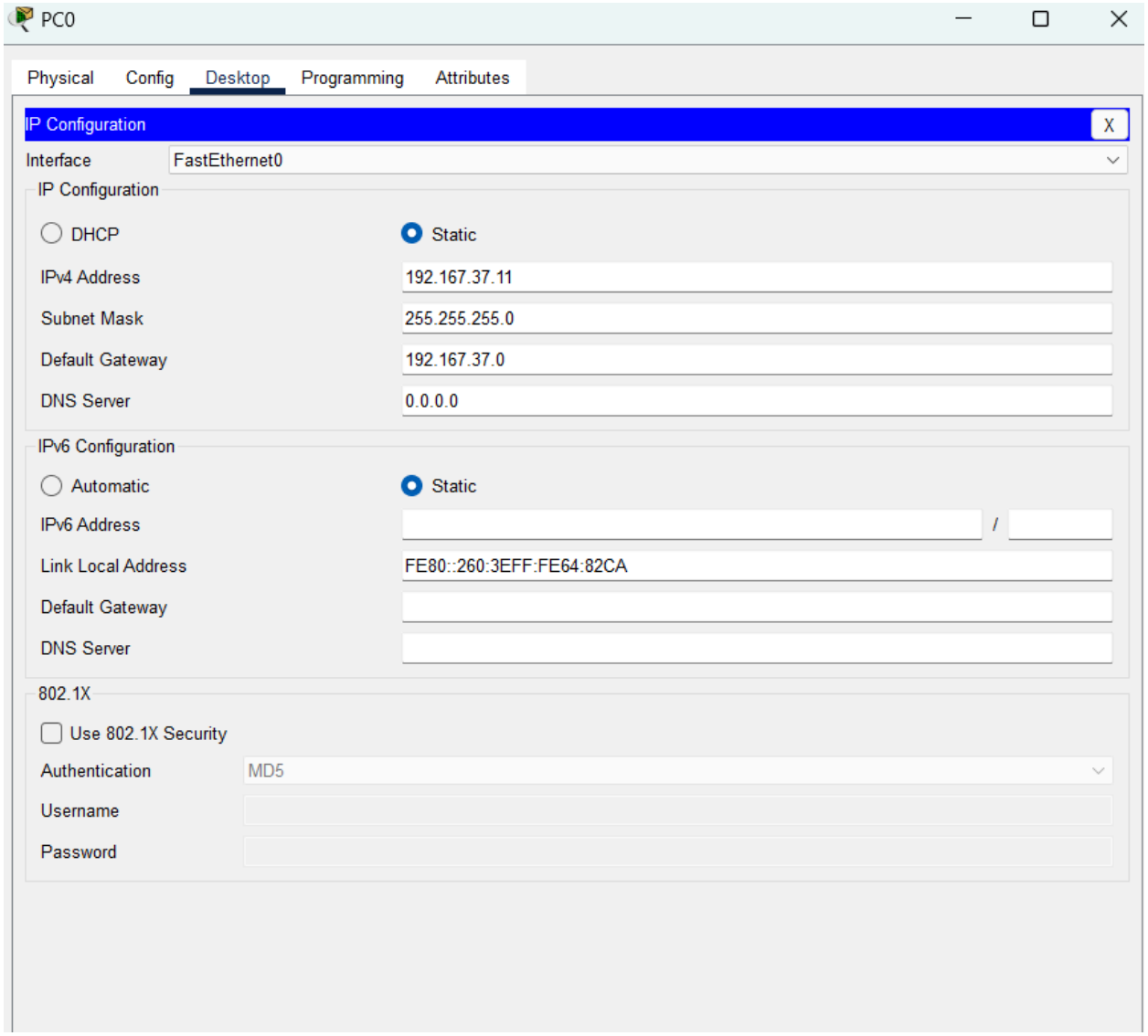

Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch#
Switch#
Switch#
Switch#
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fast
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface fast
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface fast
Switch(config)#interface fastEthernet 0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#
Switch(config)#interface fast
Switch(config)#interface fastEthernet 0/4
Switch(config-if)#switchport mode trunk
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
Switch(config-if)#switchport trunk allowed vlan all
Switch(config-if)#no shutdown
Switch(config-if)#
Switch(config-if)#
Copy
Paste
Top

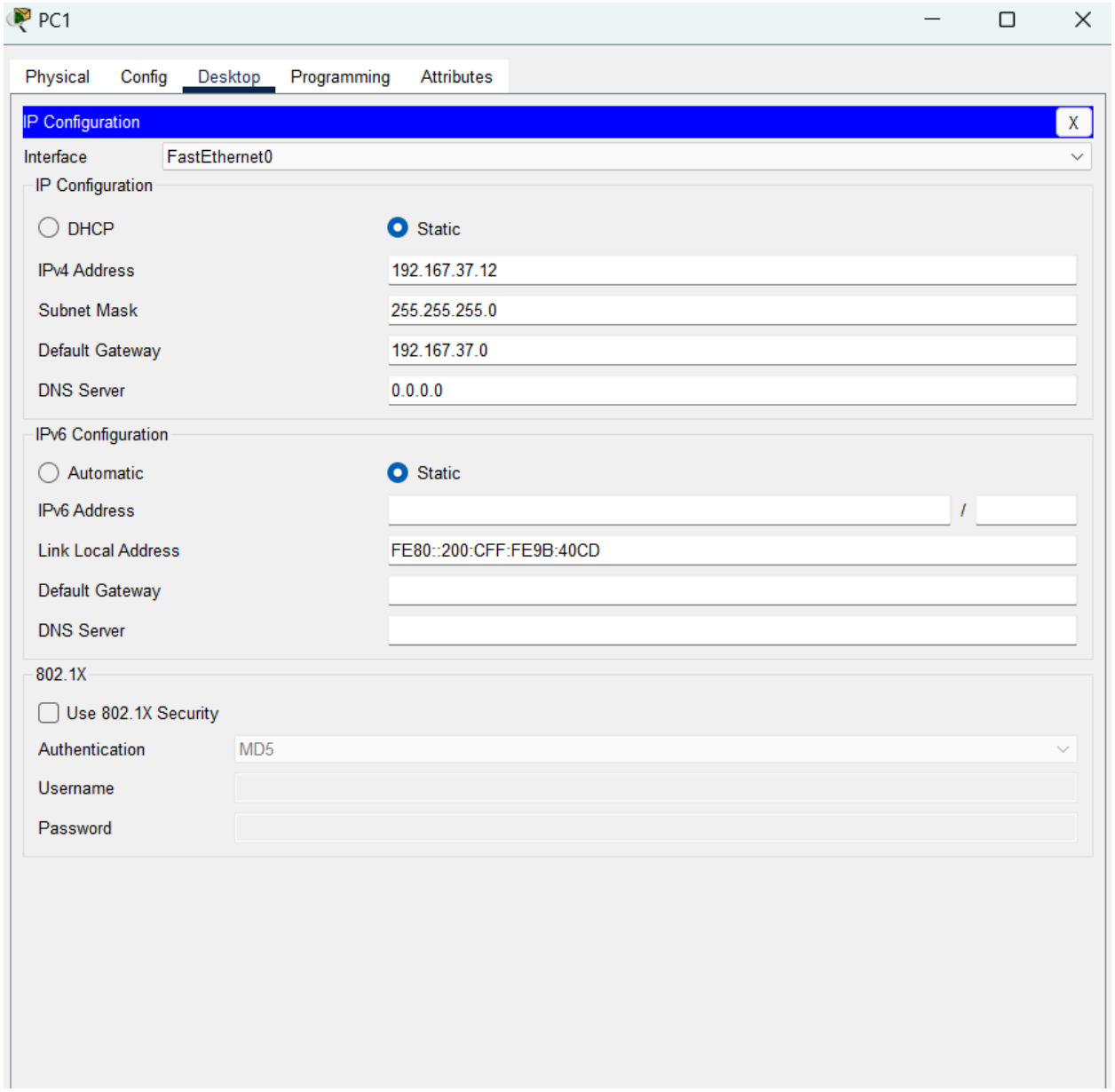

Switch1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fast
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface fast
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface fast
Switch(config)#interface fastEthernet 0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fast
Switch(config)#interface fastEthernet 0/4
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan all
Switch(config-if)#
Switch(config-if)#no shutdown
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface fast
Switch(config)#interface fastEthernet 0/5
Switch(config-if)#switchport mode trunk
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
Switch(config-if)#switchport trunk allowed vlan all
Switch(config-if)#
Switch(config-if)#no shutdown
Switch(config-if)#exit
Copy
Paste
Top

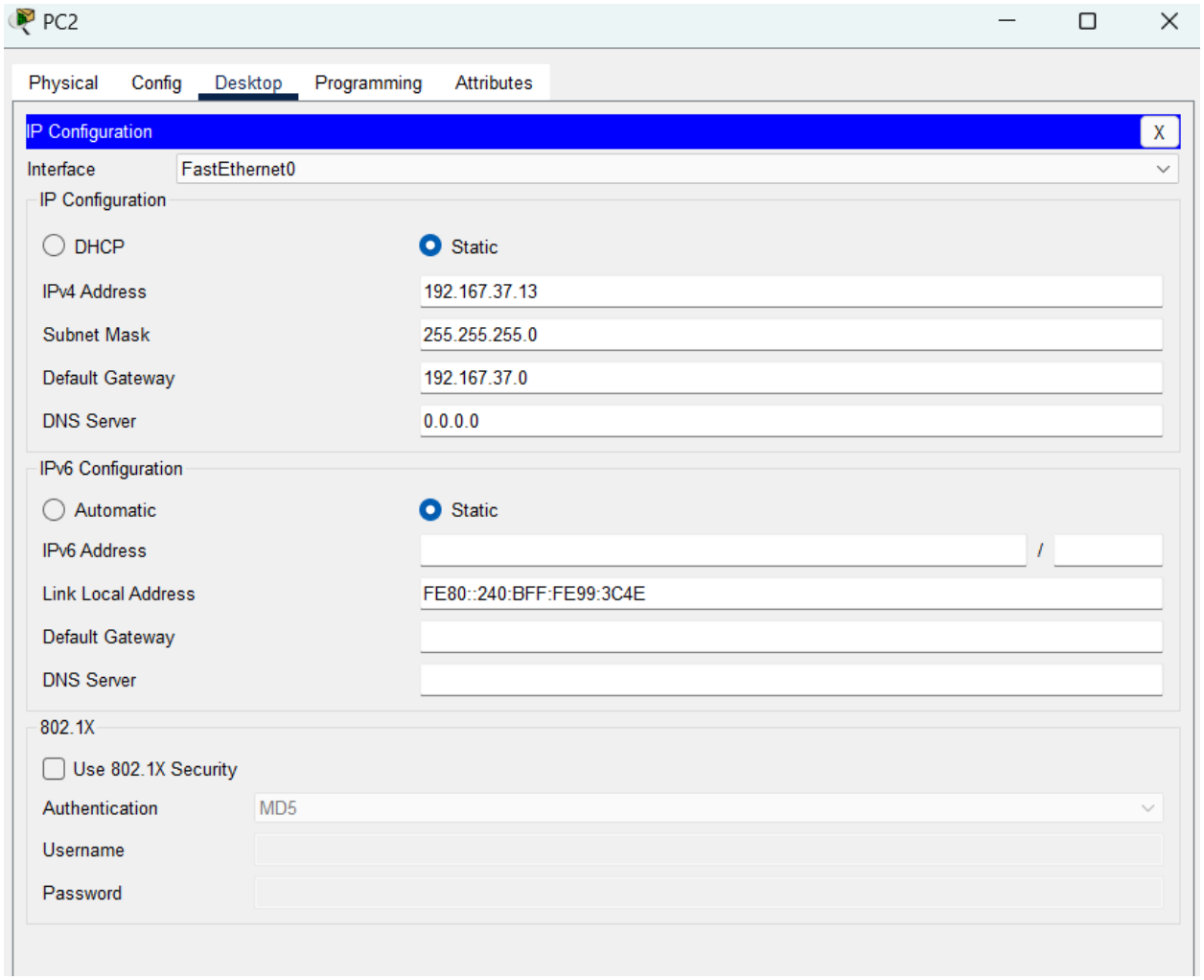


Step 2: Setting Up IPs

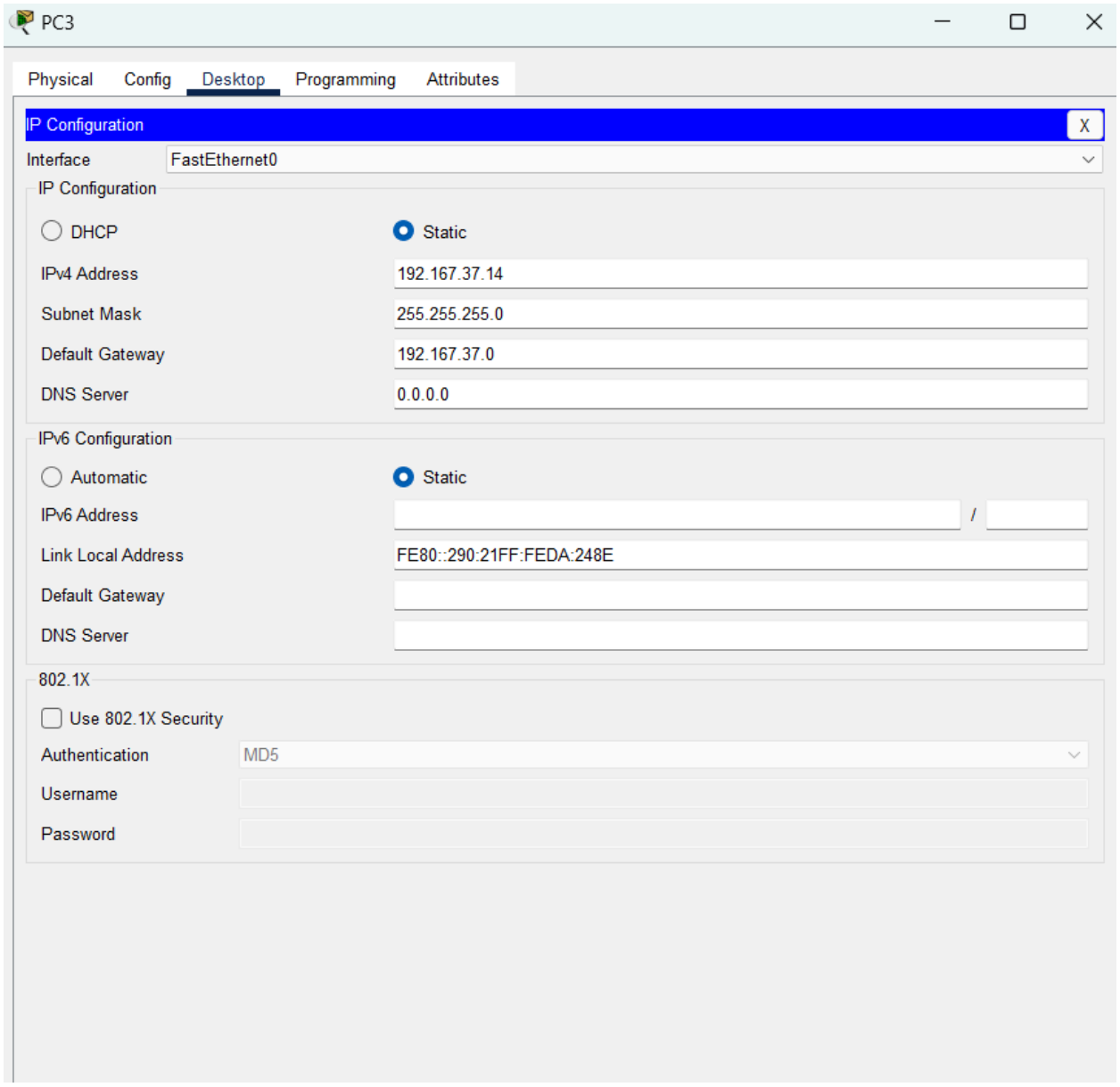
VLAN 10 host's IP address for three pc

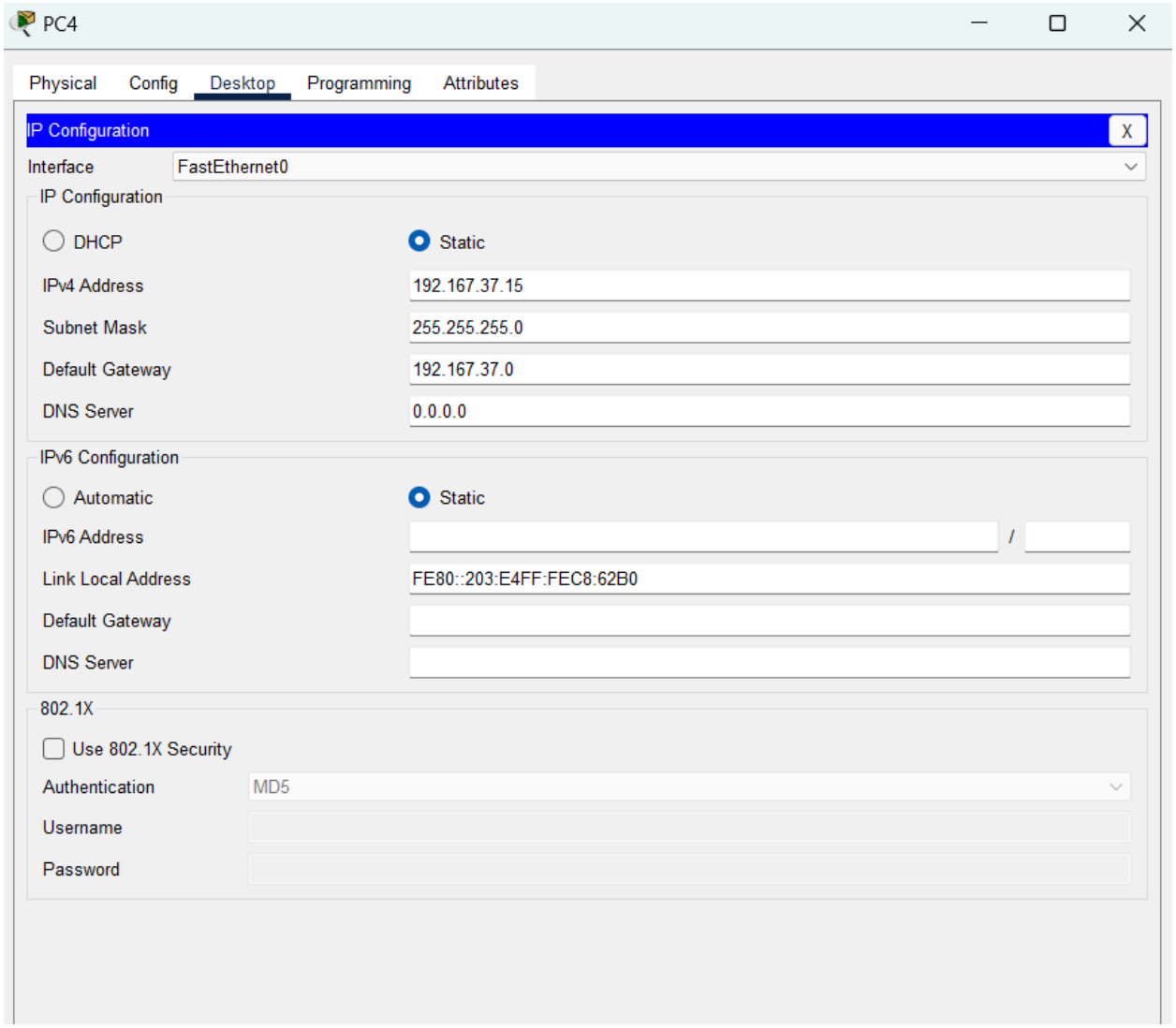

PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.167.37.11
Subnet Mask
255.255.255.0
Default Gateway
192.167.37.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::260:3EFF:FE64:82CA
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

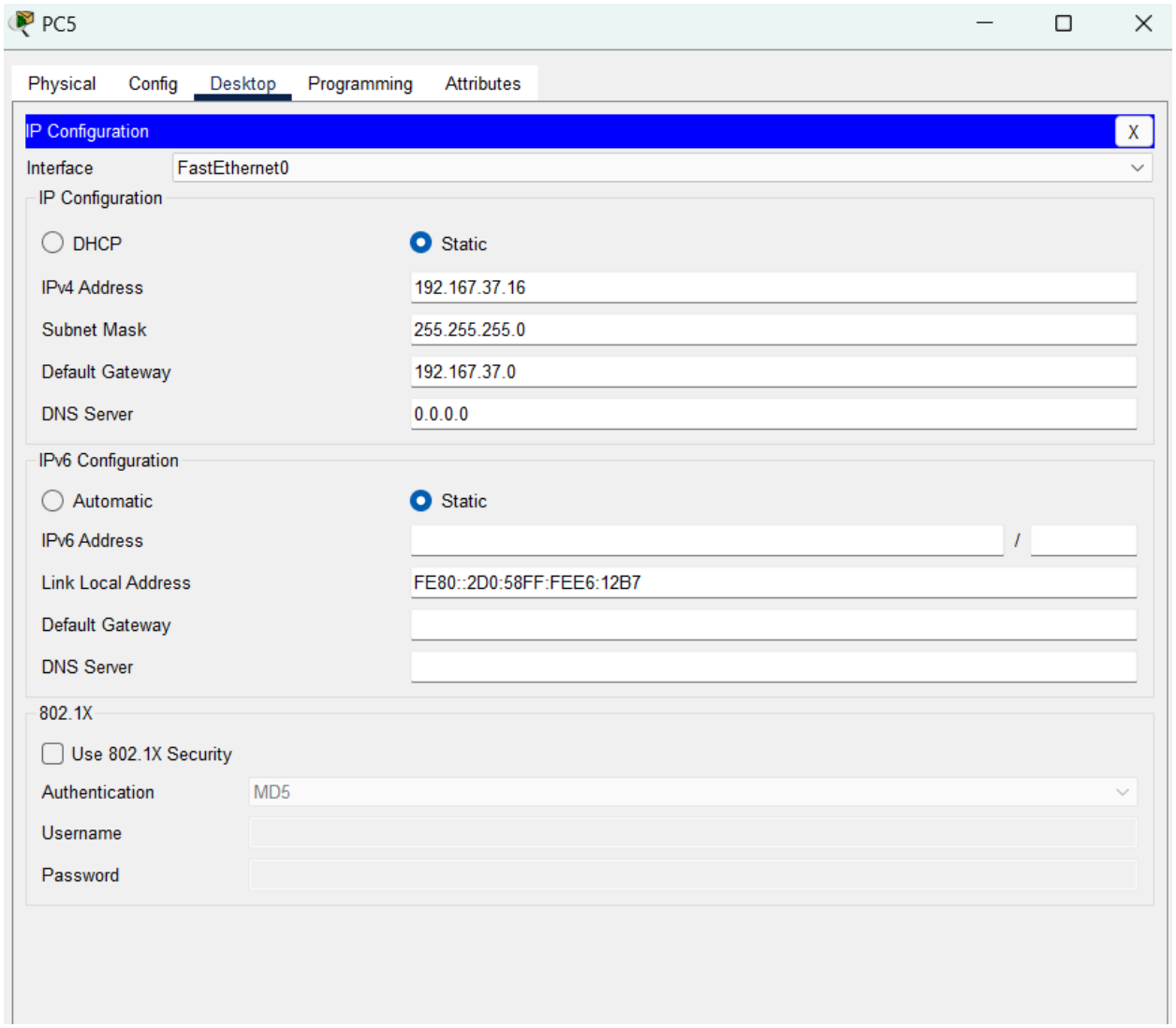

PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.167.37.12
Subnet Mask
255.255.255.0
Default Gateway
192.167.37.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::200:CFF:FE9B:40CD
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password



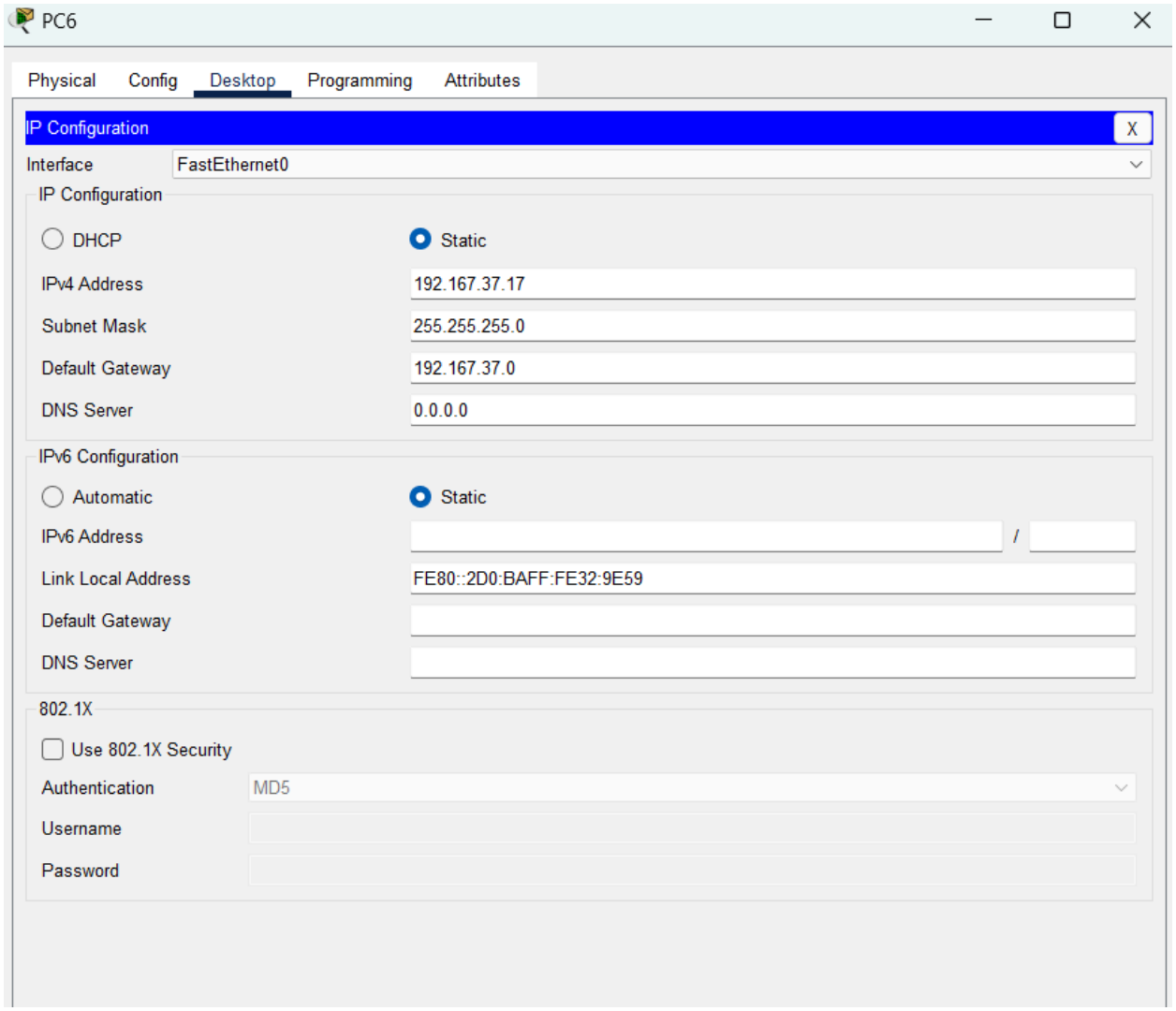
VLAN 20 host’s IP address for three pc

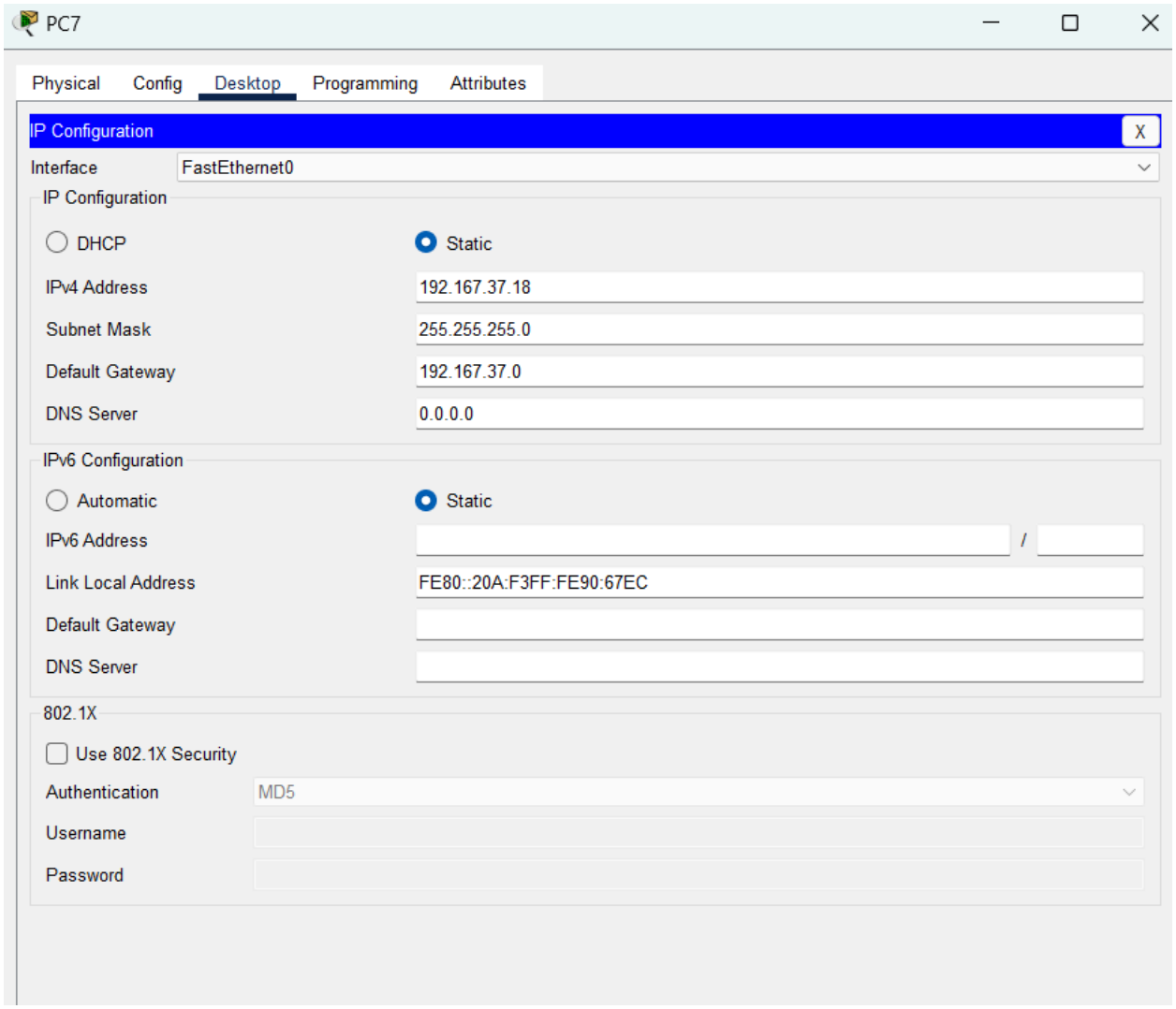

PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.167.37.14
Subnet Mask
255.255.255.0
Default Gateway
192.167.37.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::290:21FF:FEDA:248E
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

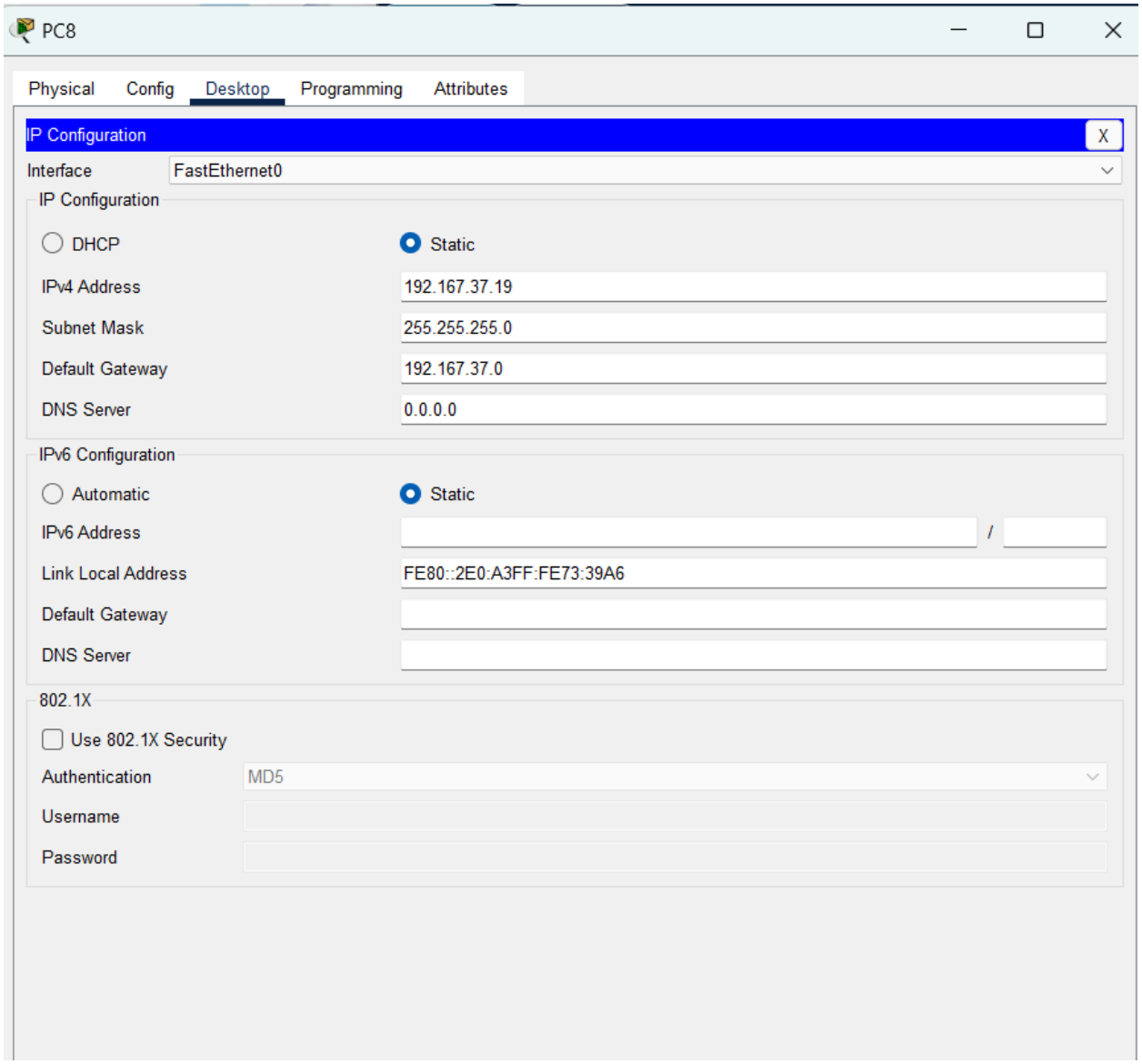

PC4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.167.37.15
Subnet Mask
255.255.255.0
Default Gateway
192.167.37.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::203:E4FF:FEC8:62B0
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password



VLAN 30 host's IP address for three pc


PC6
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.167.37.17
Subnet Mask
255.255.255.0
Default Gateway
192.167.37.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::2D0:BAFF:FE32:9E59
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password


PC7
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.167.37.18
Subnet Mask
255.255.255.0
Default Gateway
192.167.37.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::20A:F3FF:FE90:67EC
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password



Questions (Answer to the point):

1. How many host bits are needed in the largest required subnet?
   Ans: 6 bits.
2. How many VLANs need to be configured to each of the switches?
   Ans: 3 VLANs.
3. Which interfaces need Access Link?
   Ans: Interfaces that are connected with PCs.
4. Which interfaces need Trunk Link?
   Ans:Interfaces that are connected with another switch.
5. After configuring VLAN, what will happen if we broadcast?

Ans: After configuring VLANs, broadcast traffic will only be forwarded within the same VLAN, limiting its scope.